PROJECT AND TEAM INFORMATION

Project Title

SMART AMBULANCE ROUTE AND TRAFFIC OPTIMIZER

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

During emergency situations, ambulance has to reach the emergency location and hospital in the fastest way possible. Congestion on the roads, roadblocks, traffic, and inaccessible roads increases the travel time. The proposed Smart Ambulance Route Traffic Optimizer will have a graphical simulation for selecting the suitable ambulances, finding hospitals, and shortest path using Data Structures and Algorithms. The project will demonstrate the application of C programming, DSA, and graphical programming to solve a real-life problem. The algorithms such as Dijkstra, BFS, and DFS will be implemented for route finding.

State of the Art / Current solution (1 pt)

Current navigation systems depend on GPS and traffic data to find the paths.. These systems need an internet connection and rely on external services.The proposed project will show a simulation built using the C programming language and the raylib graphics library.The city will be represented as a weighted graph. Each road will carry information about distance and traffic conditions. The simulation will display ambulances, hospitals, emergency priority levels, traffic conditions and road accidents or blockages.

The following features will be demonstrated in the proposed project:

- *Ambulance selection*
- *Hospitals listing*
- *Emergency severity and priority*
- *Dynamic traffic updates*
- *Road accidents or blockages*
- *Automatic route recalculation*
- *Shortest versus traffic-optimized routes*
- *Alternative route suggestions*
- *BFS/DFS visualization*
- *Ambulance movement animation*
- *Ambulance status updates*
- *Route history tracking*
- *Simulation reset*

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Project Goals and Milestones (2 pts)

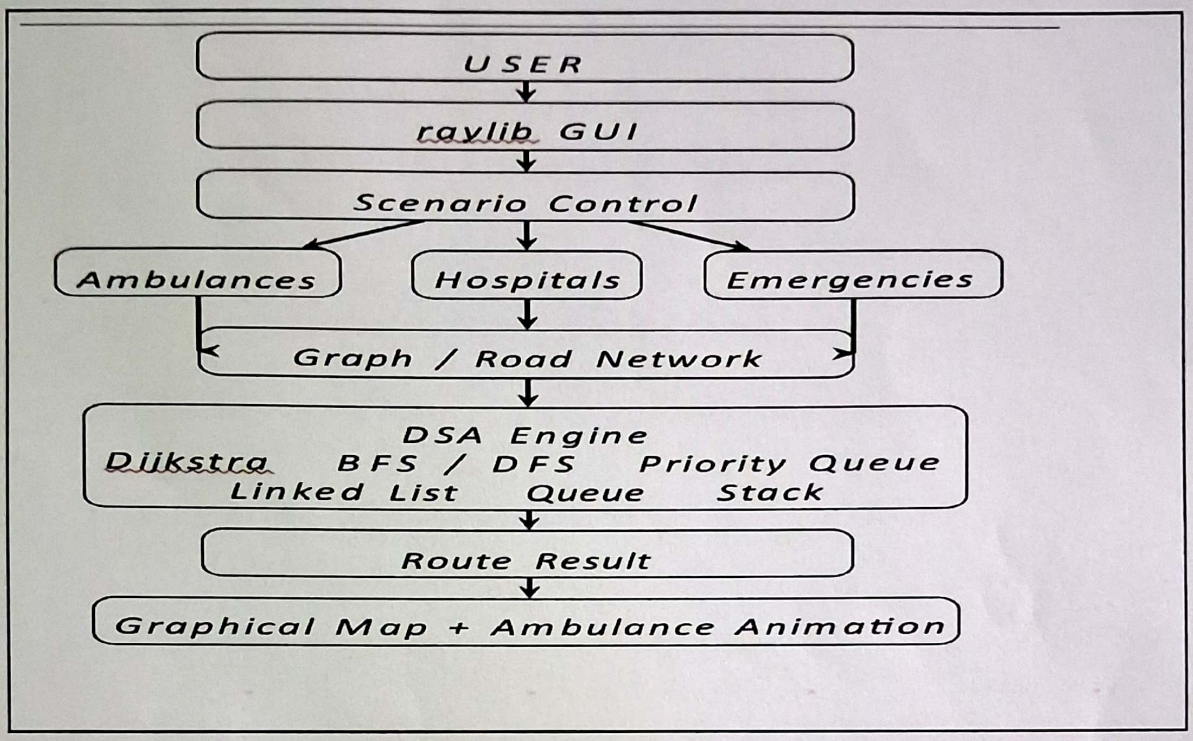
1. *Problem Statement: Identify ambulance routing challenges and determine the required features.*
2. *System Design: Create data structures for the graph, ambulance, hospital, road and emergency.*
3. *DSA Implementation: Build arrays, linked lists, queues, stacks and graph representations.*
4. *Algorithm Implementation: Implement BFS, DFS, priority queues and Dijkstras algorithm.*
5. *GUI Development: Develop the city map using raylib.*
6. *Feature Interaction: Add traffic conditions, road blockages, multiple ambulances and hospitals.*
7. *Testing: Check routing accuracy, road closures, traffic data, priorities and connectivity.*
8. *Final Demonstration : Present the ambulance routing simulation.*

Project Approach (3 pts)

The Smart Ambulance Route Traffic Optimizer will be developed in C with the raylib graphics library. The program will show the city as a weighted graph. Intersections will be nodes and roads will be edges with each edge storing distance, traffic data and other details. Ambulances and hospitals will be stored in structures using arrays. When an emergency occurs the system The Smart Ambulance Route Traffic Optimizer will be developed in C with the raylib graphics library. The program will show the city as a weighted graph. Intersections will be nodes and roads will be edges with each edge storing distance, traffic data and other details. Ambulances and hospitals will be stored in structures using arrays. When an emergency occurs the system .

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System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The final product will be a graphical ambulance routing simulation created in C using the raylib library. The simulation will feature:

- *Interactive city road map*
- *Several ambulances and hospitals*
- *Prioritization of emergency calls*
- *Traffic simulation*
- *Road closure and reopening*
- *Optimization based on Dijkstra Algorithm*
- *Recomputation of routes*
- *Breadth First Search(BFS) and Deapth First Search (DFS) Algorithms*
- *Ambulance animation*
- *Comparison of alternate route/paths*
- *Route statistics*
- *History*
- *Reset*

The project will demonstrate the practical applicability of Data Structures and Algorithms in a C programming language to solve a real-world problem.

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Assumptions

The project will employ a simulated city map and the distance and traffic data for the roads will be static. Traffic simulation will be manually controlled by a user. Ambulances and hospitals will be represented as nodes in the graph. The system will not make use of actual GPS data or traffic data. The application will not require an internet connection. The project will be a simulation made for educational purposes; therefore, no actual emergency vehicles or communication systems will be integrated. All the algorithms required for this project will be implemented in C programming language. The graphical interface of the application will be built using C + raylib.

References

1. *Thomas H. Cormen et al., Introduction to Algorithms , MIT Press.*
2. *Mark Allen Weiss, Data Structures and Algorithm Analysis .*
3. *Robert Lafore, Data Structures and Algorithms .*
4. *Raylib online documentation and learning resources.*
5. *C programming language online documentation and learning resources.*

Shaurti
11-09-26

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